

FedCDKD: Conflict-Driven Knowledge Distillation for Heterogeneous Federated Visual Recognition

Yuanshuo Lin  and Yong Wang^(✉) 

College of Computer Science and Technology, Harbin Engineering University, Harbin, China

✉ wangyongcs@hrbeu.edu.cn

Abstract. Federated visual recognition in practice often combines non-independent and identically distributed (non-IID) image streams, heterogeneous neural backbones, and intermittent client participation. These factors can make client update directions disagree, which weakens standard aggregation and makes fixed-strength knowledge transfer brittle. To this end, we propose **FedCDKD** (Conflict-Driven Knowledge Distillation for Heterogeneous Federated Learning), a closed-loop framework that uses aggregation conflict measured on the server to regulate local mutual distillation. The server tracks client update directions with EMA, solves a min-norm quadratic program for a conflict-suppressed consensus update, and broadcasts a client-specific conflict degree. Each client uses this scalar to adjust the temperature and the weights of the supervised loss and KL divergence loss during distillation between the shared aggregatable model and the private personalized model. Experiments on CIFAR-10, CIFAR-100, and Tiny-ImageNet demonstrate that FedCDKD consistently delivers the best performance across diverse data partitions, model heterogeneity levels, and participation ratios, showing strong robustness and stable gains under challenging federated settings.

Keywords: Federated Learning · Conflict-Aware Optimization · Model Heterogeneity · Statistical Heterogeneity · Partial Participation · Knowledge Distillation

1 Introduction

Federated learning (FL) supports privacy-preserving visual recognition on distributed edge devices by training models without centralizing raw images [1]. Real deployments, however, rarely match the clean setting assumed by standard FL algorithms. Clients may observe different categories or scene statistics, use different backbones under hardware constraints, and participate only when network and energy conditions permit. When data heterogeneity, model heterogeneity, and partial participation coexist, client updates can disagree sharply in direction, which makes aggregation unstable and degrades accuracy [2,3,4,5]. Figure 1 shows that the interaction between data and model heterogeneity can introduce additional accuracy decay beyond either factor alone.

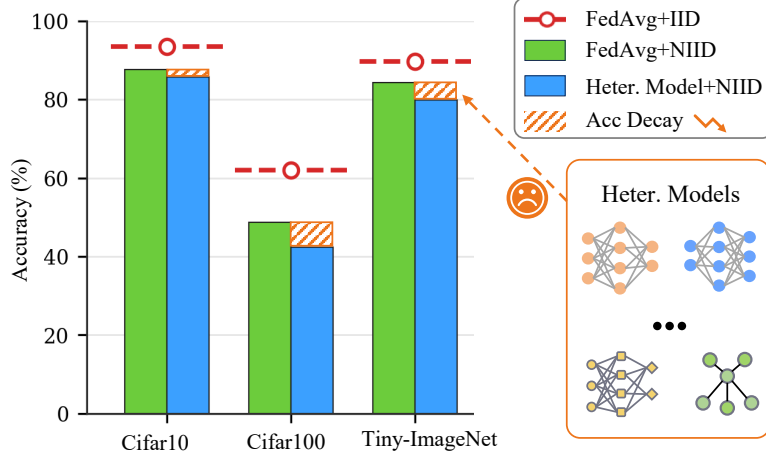


Fig. 1. Interaction between data and model heterogeneity in federated visual recognition. The combined setting introduces additional accuracy decay, motivating conflict-aware aggregation and adaptive knowledge transfer.

Existing methods cover only part of this setting. Optimization-based FL can reduce non-IID drift [3,4,5,6,7], but it usually assumes that all clients share the same architecture. Heterogeneous FL avoids direct parameter matching through prototypes[8], shared heads[9], or distillation [10,11], but the transfer strength is often fixed or tuned separately from server aggregation. This separation becomes brittle under partial participation, where the active client set and the conflict pattern change from round to round. A fixed distillation weight may be too weak for compatible clients yet harmful for strongly misaligned clients. Heterogeneous federated visual recognition therefore needs to measure update conflict and adapt cross-model transfer within the same training loop.

We propose **FedCDKD** (Conflict-Driven Knowledge Distillation for Heterogeneous Federated Learning), a framework that treats update conflict as a measurable control signal to close the loop between server aggregation and client adaptation. Instead of treating conflict-aware aggregation and distillation-based transfer as isolated processes, FedCDKD bridges these two paradigms through a unified feedback mechanism. Specifically, the server tracks client update directions using an exponential moving average (EMA) and solves a min-norm quadratic program to achieve a conflict-suppressed consensus. This process yields a client-specific conflict degree, $c_i^t \in [0, 1]$. Each client then uses this scalar to dynamically adapt the mutual distillation between its shared aggregatable model θ and private personalized model ϕ . Unlike traditional methods that rely on fixed or manually tuned transfer strengths, FedCDKD utilizes the server-side conflict measurement to directly control the next-round client-side distillation: high conflict yields softer, more conservative transfer, while low conflict permits

stronger mutual distillation. Notably, this dynamic regulation is highly efficient, requiring the exchange of only θ and a single scalar.

The main contributions are as follows:

1. **Problem Formulation & Framework Design:** Identification of update-direction conflict as a common failure mode in heterogeneous federated visual recognition, and the introduction of a closed-loop framework that links conflict-suppressed server aggregation with conflict-adaptive client distillation.
2. **Coupled Server-Client Mechanism:** The design of a unified feedback loop where a server-side min-norm quadratic program tracks EMA directions to extract a personalized conflict signal, which directly adapts the temperature and loss weights for client-side mutual distillation between a shared aggregatable model and a private personalized model.
3. **Extensive Empirical Validation:** Comprehensive evaluation of the proposed method on CIFAR-10, CIFAR-100, and Tiny-ImageNet, confirming its effectiveness under the joint presence of non-IID data partitions, heterogeneous backbones, and partial participation.

2 Related Work

2.1 Federated Optimization and Aggregation Conflicts

Data heterogeneity is a major source of instability in FL because non-IID data produces biased local optima and inconsistent update directions. FedAvg [1] is the standard baseline, while FedProx[3], SCAFFOLD[4], FedDyn[5], FedNova[12], and FedDC[13] reduce drift through regularization[3], control variates[4], dynamic terms[12], normalized updates[5], or drift correction[13]. Recent aggregation methods further show that server updates should account for client disagreement instead of simply averaging all updates, e.g., FedLAW[7], FedDisco[6], FedAWA[14], L-DAWA[15], FedLAMA[16], and FedCALM [17]. These methods motivate conflict-aware aggregation, but most still rely on an aggregatable common architecture or use conflict mainly to revise the server update. FedCDKD keeps the server-side benefit of conflict suppression and further reuses the measured conflict as a client-side control signal for heterogeneous distillation.

2.2 Heterogeneous Federated Learning and Knowledge Transfer

Model heterogeneity arises when clients choose different backbones because of hardware budgets, latency constraints, or local deployment needs. Since private heterogeneous models cannot be directly averaged, existing methods often use parameter-agnostic transfer: FedProto[8] and FedTGP[11] align class-level representations or global prototypes, FedGH shares a common prediction interface [9], FedKTL [18] decouples heterogeneous backbones through a target embedding space, and distillation-based methods such as FD[19], FedDF[10], FedGen[20],

FedKD[21], and FedMRL[22] transfer logits or auxiliary knowledge. These designs enable collaboration without sharing raw data or private model parameters. However, their transfer strength is usually fixed or tuned independently from the round-wise aggregation geometry. Under partial participation, the active client set and conflict pattern change across rounds; a static distillation weight can over-regularize misaligned clients or underuse compatible peers. FedCDKD addresses this gap by linking server-measured update conflict with client-specific, round-dependent mutual distillation.

3 Preliminaries

We consider an FL system with N clients. Client i holds local data D_i with $n_i = |D_i|$ samples and local risk

$$F_i(\boldsymbol{\theta}) = \frac{1}{n_i} \sum_{(\mathbf{x}, y) \in D_i} \ell(f(\mathbf{x}; \boldsymbol{\theta}), y). \quad (1)$$

At round t , the server samples $\mathcal{S}^t$ with $K = |\mathcal{S}^t|$, broadcasts $\boldsymbol{\theta}^t$, and receives local models $\boldsymbol{\theta}_i^{t+1}$ after E Stochastic Gradient Descent (SGD) steps. Standard FL aggregates them as

$$\boldsymbol{\theta}^{t+1} = \sum_{i \in \mathcal{S}^t} p_i^t \boldsymbol{\theta}_i^{t+1}, \quad p_i^t = \frac{n_i}{\sum_{j \in \mathcal{S}^t} n_j}. \quad (2)$$

This protocol becomes fragile when data heterogeneity, model heterogeneity, and partial participation coexist: local updates may point in conflicting directions, while private client backbones cannot be directly averaged.

FedCDKD therefore separates a shared aggregatable model $\boldsymbol{\theta}$ from a private personalized model $\boldsymbol{\phi}$. The two models exchange information through knowledge distillation (KL) [23]. For logits $\mathbf{z}_a, \mathbf{z}_b$ and temperature τ , let $\mathbf{p}_a = \text{softmax}(\mathbf{z}_a/\tau)$ and $\mathbf{p}_b = \text{softmax}(\mathbf{z}_b/\tau)$. A typical distillation term combines cross-entropy and KL regularization,

$$\mathcal{L}_a = \alpha \text{CE}(\mathbf{z}_a, y) + (1 - \alpha) \tau^2 \text{KL}(\mathbf{p}_b \| \mathbf{p}_a), \quad (3)$$

with an analogous objective for model b . The proposed method uses server-measured update conflict to adapt these distillation parameters on each client.

4 Methodology

FedCDKD addresses non-IID data, heterogeneous client models, and partial participation by coupling server-side conflict suppression with client-side adaptive distillation. Its main design is a closed feedback loop. The server estimates how each client update aligns with a conflict-suppressed consensus direction, and the resulting scalar conflict degree controls that client’s next-round distillation behavior.

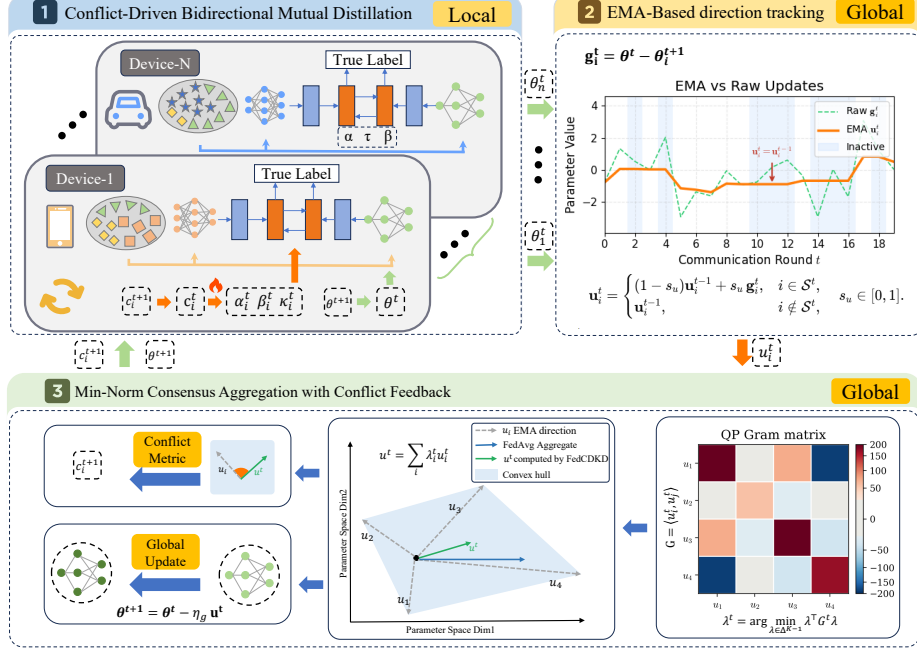


Fig. 2. Overview of FedCDKD. (1) *Conflict-driven bidirectional mutual distillation* on each client jointly updates a shared *aggregatable* model and a private *personalized* model, with distillation strength scheduled by the conflict signal. (2) *EMA-based direction tracking* on the server smooths noisy client updates and provides robust direction estimates under partial participation. (3) *Min-norm consensus aggregation with conflict feedback* computes a conflict-suppressed consensus update to refine the global model and then feeds back personalized conflict degrees for the next round.

4.1 Overview

Each client maintains a shared model θ_i with an aggregatable architecture and a private personalized model ϕ_i chosen from the local heterogeneous model pool. Only θ_i is uploaded. This is a proxy-model heterogeneous FL setting; the shared branch enables aggregation, while the private branch preserves deployment-specific architectures rather than requiring all clients to deploy the same final model. During local training, θ_i and ϕ_i learn from labels and exchange soft predictions through mutual distillation. After aggregation, the server broadcasts the updated global model and a client-specific conflict degree. High conflict makes the next local distillation softer and more conservative; low conflict permits stronger cross-model transfer.

Figure 2 illustrates the complete closed loop between client-side mutual distillation and server-side conflict feedback.

At round t , selected clients receive θ^t and cached conflict degrees, update (θ_i, ϕ_i) locally, and upload θ_i^{t+1} . The server tracks uploaded update directions,

Algorithm 1: FedCDKD: Conflict-Driven Knowledge Distillation for Heterogeneous FL

Input: client datasets $\{D_i\}_{i=1}^N$, rounds T , local epochs E , local LR η , global LR η_g , EMA coefficient s_u

Output: global model θ^T and client personalized models $\{\phi_i^T\}_{i=1}^N$

- 1 Initialize global model θ^0 ; each client initializes a personalized model ϕ_i^0
- 2 Set $\mathbf{u}_i^0 \leftarrow \mathbf{0}$ and $c_i^0 \leftarrow 0$ for all clients i
- 3 **for** $t \leftarrow 0$ **to** $T - 1$ **do**
- 4 Server samples participating clients $\mathcal{S}^t$ and broadcasts $(\theta^t, \{c_i^t\}_{i \in \mathcal{S}^t})$
- 5 **foreach** clients $i \in \mathcal{S}^t$ **in parallel do**
- 6 Set shared model $\theta_i^t \leftarrow \theta^t$
- 7 Compute $(\tau_i^t, \alpha_i^t, \beta_i^t)$ by Eq. (11)
- 8 Update (ϕ_i^t, θ_i^t) by Eq. (12) and Eq. (13)
- 9 Upload θ_i^{t+1} to server
- 10 **foreach** clients $i \in \mathcal{S}^t$ **do**
- 11 Compute $\mathbf{g}_i^t$ by Eq. (4) and update $\mathbf{u}_i^t$ by Eq. (5)
- 12 Solve Eq. (6) to get λ^t , compute $\mathbf{u}^t$ by Eq. (7), and update θ^{t+1} by Eq. (8)
- 13 Compute and cache $\{c_i^{t+1}\}$ by Eq. (10)
- 14 **return** θ^T and $\{\phi_i^T\}_{i=1}^N$

solves a min-norm consensus problem, updates θ^{t+1} , and computes conflict degrees for the next round. Algorithm 1 summarizes the procedure.

4.2 Conflict-Suppressed Server Aggregation

For a participating client $i \in \mathcal{S}^t$, the raw update direction is

$$\mathbf{g}_i^t = \theta^t - \theta_i^{t+1}. \quad (4)$$

To reduce round noise and handle partial participation, the server maintains an EMA direction

$$\mathbf{u}_i^t = \begin{cases} (1 - s_u)\mathbf{u}_i^{t-1} + s_u \mathbf{g}_i^t, & i \in \mathcal{S}^t, \\ \mathbf{u}_i^{t-1}, & i \notin \mathcal{S}^t, \end{cases} \quad (5)$$

where $s_u \in [0, 1]$. Given the active EMA directions, FedCDKD computes a conflict-suppressed consensus by solving

$$\lambda^t = \arg \min_{\lambda \in \Delta^{K-1}} \left\| \sum_{i \in \mathcal{S}^t} \lambda_i \mathbf{u}_i^t \right\|_2^2 = \arg \min_{\lambda \in \Delta^{K-1}} \lambda^\top \mathbf{G}^t \lambda, \quad (6)$$

where Δ^{K-1} is the probability simplex and $G_{ij}^t = \langle \mathbf{u}_i^t, \mathbf{u}_j^t \rangle$. This min-norm formulation is closely related to multi-objective conflict mitigation [24, 25]: directions that oppose the consensus tend to receive small weights. The consensus direction and global update are

$$\mathbf{u}^t = \sum_{i \in \mathcal{S}^t} \lambda_i^t \mathbf{u}_i^t. \quad (7)$$

$$\boldsymbol{\theta}^{t+1} = \boldsymbol{\theta}^t - \eta_g \mathbf{u}^t. \quad (8)$$

The supplementary material gives the KKT interpretation and complexity details.

FedCDKD then converts the alignment between each client direction and the consensus into a bounded conflict degree. By default, we use the cosine-inconsistency metric:

$$\text{sim}_i^t = \frac{\langle \mathbf{u}_i^t, \mathbf{u}^t \rangle}{\|\mathbf{u}_i^t\|_2 \|\mathbf{u}^t\|_2 + \varepsilon}, \quad \mathcal{M}_{\cos}(\mathbf{u}_i^t, \mathbf{u}^t) = \frac{1 - \text{sim}_i^t}{2}, \quad (9)$$

where ε avoids numerical instability when a direction norm is close to zero. The broadcast conflict signal is

$$c_i^{t+1} = \text{clip}(\mathcal{M}_{\cos}(\mathbf{u}_i^t, \mathbf{u}^t), 0, 1). \quad (10)$$

Thus, better alignment with the consensus gives smaller conflict, while orthogonal or opposite directions induce larger conflict. The supplement also reports angle- and aggregation-weight-based variants to test metric sensitivity. Only this scalar and the shared model are broadcast. The scalar does not expose raw data, labels, logits, or the private model. Repeated broadcasts may still reveal coarse alignment trends, so clipping, quantization, delay, or secure aggregation can be used in privacy-sensitive deployments.

4.3 Conflict-Driven Mutual Distillation

Client i maps c_i^t to the temperature and task-loss weights used by the two local models:

$$\begin{cases} \tau_i^t = \tau_{\text{low}} + (\tau_{\text{high}} - \tau_{\text{low}})(c_i^t)^{\gamma_\tau}, \\ \alpha_i^t = \alpha_{\text{high}} + (\alpha_{\text{low}} - \alpha_{\text{high}})(1 - c_i^t)^{\gamma_\alpha}, \\ \beta_i^t = \beta_{\text{high}} + (\beta_{\text{low}} - \beta_{\text{high}})(1 - c_i^t)^{\gamma_\beta}. \end{cases} \quad (11)$$

With $\alpha_{\text{low}} < \alpha_{\text{high}}$ and $\beta_{\text{low}} < \beta_{\text{high}}$, larger conflict increases temperature and supervised weights, reducing reliance on KL matching when the two models are likely misaligned. Lower conflict keeps stronger mutual distillation. This schedule is a bounded design choice rather than a separate claimed contribution; sensitivity to the conflict metric and EMA coefficient is evaluated in Section 5.4 and the supplement.

For a sample $(\mathbf{x}, y)$, let $\mathbf{z}_p = f(\mathbf{x}; \boldsymbol{\phi}_i^t)$ and $\mathbf{z}_s = f(\mathbf{x}; \boldsymbol{\theta}_i^t)$. With $\mathbf{p}_p = \text{softmax}(\mathbf{z}_p / \tau_i^t)$ and $\mathbf{p}_s = \text{softmax}(\mathbf{z}_s / \tau_i^t)$, FedCDKD optimizes

$$\mathcal{L}_p^i = \alpha_i^t \text{CE}(\mathbf{z}_p, y) + (1 - \alpha_i^t)(\tau_i^t)^2 \text{KL}(\mathbf{p}_s \| \mathbf{p}_p), \quad (12)$$

$$\mathcal{L}_s^i = \beta_i^t \text{CE}(\mathbf{z}_s, y) + (1 - \beta_i^t)(\tau_i^t)^2 \text{KL}(\mathbf{p}_p \| \mathbf{p}_s). \quad (13)$$

After local training, only the shared model $\boldsymbol{\theta}_i^{t+1}$ is uploaded, while $\boldsymbol{\phi}_i^{t+1}$ remains private.

5 Experiments

We evaluate FedCDKD under the simultaneous presence of model heterogeneity, data heterogeneity, and partial participation. To assess its effectiveness, we consider the following research questions (RQs):

RQ 1 *How well does FedCDKD improve performance across different datasets and data partitions?*

RQ 2 *How robust is FedCDKD under varying levels of client participation and model heterogeneity?*

RQ 3 *What is the contribution of each major component in FedCDKD?*

5.1 Experimental Setup

Datasets and metric. The main evaluation uses CIFAR-10/100 [26] and Tiny-ImageNet [27]; Unless otherwise stated, we report the average test accuracy of the deployed personalized models ϕ . We evaluate IID and non-IID splits including $\text{Dir}(0.1)$, $\text{Dir}(0.3)$, and $\text{Pat}(2)$ [1]. Because evaluation is performed on deployed personalized models and client-wise test distributions, the partition labels indicate data-generation protocols rather than guaranteeing a monotonic ordering of accuracy across all datasets.

Model heterogeneity. Clients draw private backbones from heterogeneous visual model pools, including ResNet variants [28], MobileNetV2 [29], GoogLeNet [30], and lightweight CNNs [31]. We vary the number of candidate model families $M \in \{2, 3, 4, 5, 8, 9\}$ to control heterogeneity strength. Each client deploys its private ϕ , while θ uses a common aggregatable architecture for communication. This setting should be understood as heterogeneous personalization with a shared proxy branch, not as fully architecture-free aggregation. Detailed splits, model pools, and preprocessing are provided in the supplement.

5.2 RQ1: Effectiveness across Datasets and Partitions

Main results across visual datasets and partitions. Table 1 reports accuracy under four partitions on the visual recognition datasets. FedCDKD achieves the best accuracy across all visual datasets and partitions. The gains are clearer in harder regimes, such as severe label skew and higher-class visual recognition, where objective mismatch and architecture diversity induce stronger conflicts. We avoid interpreting the gains as uniformly large: CIFAR-10 margins are sometimes modest, while CIFAR-100 and Tiny-ImageNet show clearer improvements. For example, under $\text{Dir}(0.1)$, FedCDKD improves over the best competing result by 1.52% on CIFAR-100 and 2.14% on Tiny-ImageNet. Under IID, where optimization is less conflict-dominated, the improvements on CIFAR-100 and Tiny-ImageNet remain 2.42% and 3.62%, respectively. These results indicate that conflict-suppressed aggregation and adaptive transfer improve heterogeneous visual recognition without amplifying negative transfer.

Table 1. Top-1 test accuracy (%) of personalized client models on visual recognition benchmarks.

Methods	CIFAR-10				CIFAR-100				Tiny-ImageNet			
	Dir(0.1)	Dir(0.3)	Pat(2)	IID	Dir(0.1)	Dir(0.3)	Pat(2)	IID	Dir(0.1)	Dir(0.3)	Pat(2)	IID
LG-FedAvg	84.94	69.84	50.93	86.35	42.35	40.21	32.17	53.51	38.37	35.80	30.76	49.97
FedGen	84.60	70.07	51.48	86.52	41.79	39.77	31.53	52.63	33.23	30.74	28.94	44.42
FedGH	85.08	68.30	52.24	85.25	41.47	40.53	30.33	49.63	37.64	34.84	28.48	50.41
FML	84.92	71.48	52.11	86.61	47.96	40.25	31.17	57.04	39.62	36.05	30.87	52.21
FedKD	84.50	70.64	51.43	85.82	43.39	41.63	34.51	53.51	38.92	34.96	29.99	51.93
FedMRL	86.93	72.07	53.67	87.02	47.93	41.43	32.74	56.35	36.82	33.72	27.84	47.72
FD	86.70	71.19	53.06	87.29	43.12	42.34	33.93	54.52	37.63	33.79	26.38	49.89
FedProto	83.65	68.41	51.02	80.35	37.95	35.80	29.90	42.54	38.03	36.84	30.85	50.95
FedTGP	86.48	70.60	51.30	89.34	46.63	45.00	33.52	54.48	40.82	37.10	32.66	52.71
FedKTL	85.75	70.82	51.45	87.49	46.94	44.84	33.68	58.42	40.60	35.83	31.73	53.12
FedCDKD	87.12	72.61	53.96	90.34	49.48	45.78	34.62	60.84	42.96	38.35	33.65	56.74

Table 2. Personalized-model accuracy (%) under partial participation on CIFAR-10 with Dir(0.1).

Join Clients	10/100	30/100	50/100	100/100
LG-FedAvg	68.43±0.94	77.84±0.69	80.02±0.34	84.94±0.27
FedGen	60.96±0.95	72.22±0.57	77.82±0.43	84.60±0.20
FedGH	75.35±0.84	77.93±0.63	81.34±0.45	85.08±0.24
FML	74.64±0.74	80.08±0.63	81.42±0.42	84.92±0.18
FedKD	72.43±0.38	74.57±0.33	78.32±0.26	84.50±0.15
FedMRL	70.53±0.31	76.74±0.31	80.20±0.16	86.93±0.10
FD	72.64±0.31	74.57±0.32	80.41±0.25	86.70±0.09
FedProto	69.30±0.42	75.31±0.30	77.50±0.21	83.65±0.14
FedTGP	74.64±0.18	77.26±0.15	79.04±0.12	86.48±0.08
FedKTL	74.74±0.20	79.36±0.12	80.16±0.12	85.75±0.10
FedCDKD	78.56±0.17	80.51±0.10	82.62±0.07	87.12±0.02

Effect of local training epochs. Figure 3 varies local epochs $E \in \{1, 3, 10\}$ on CIFAR-10 with Dir(0.1). Larger E usually increases client drift under non-IID data because local objectives are optimized for longer before aggregation. FedCDKD remains best across all three settings and reaches 91.35% at $E = 10$. The result suggests that using conflict to regulate both aggregation and distillation is useful when local updates become more aggressive.

5.3 RQ2: Robustness Under Participation and Model Heterogeneity

Partial participation. We simulate partial participation on CIFAR-10 with Dir(0.1) by fixing $N = 100$ and varying active clients $K \in \{10, 30, 50, 100\}$. As shown in Table 2, FedCDKD achieves the best mean accuracy across all ρ . Notably, FedCDKD outperforms the strongest baseline by a substantial margin of 3.21% in the highly restrictive setting of $K = 10$, while maintaining steady improvements of 0.43%, 1.20%, and 0.19% at $K = 30, 50$, and 100, respectively. Meanwhile, its reported standard deviation is at most 0.17 and drops to 0.02 under full participation. These results suggest that EMA direction tracking and cross-round conflict feedback stabilize aggregation when the active population changes.

Stronger model heterogeneity. Figure 4 increases the number of candidate model families from $M = 2$ to $M = 9$ on CIFAR-10 and CIFAR-100 with

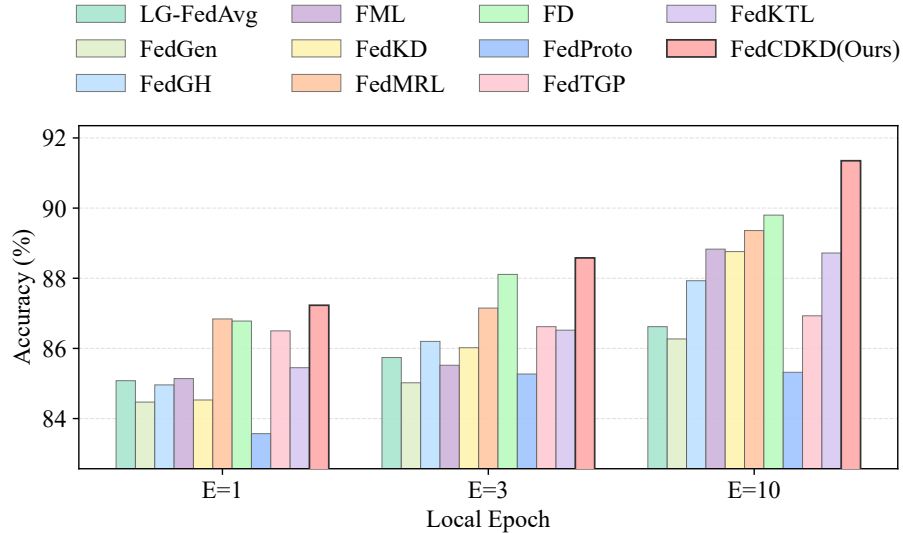


Fig. 3. Accuracy(%) of local training epochs E on CIFAR-10 with $\text{Dir}(0.1)$. Increasing E gives each client more local update steps and therefore tends to amplify client drift; FedCDKD remains stable because conflict-suppressed aggregation and adaptive distillation reduce the accumulation of conflicting updates.

Table 3. Ablation of the MinNorm aggregation and adaptive distillation components on CIFAR-10 and CIFAR-100.

MinNorm Aggregation	Adaptive Distillation	CIFAR-10	CIFAR-100
✗	✗	84.10±0.12	43.24±0.08
✗	✓	85.32±0.17	44.74±0.08
✓	✗	85.12±0.24	45.23±0.15
✓	✓	87.32±0.06	49.84±0.04

$\text{Dir}(0.1)$. All methods lose accuracy as the model pool becomes more diverse, but FedCDKD degrades more mildly and keeps the highest accuracy throughout. This trend is consistent with the motivation that fixed transfer is less reliable when clients use increasingly different architectures.

5.4 RQ3: Ablation and Component Analysis

Ablation study. Table 3 validates the contributions of two components: server-side conflict-suppressed aggregation and client-side conflict-driven mutual distillation. Each component improves over the vanilla setting, and combining them gives the best results, supporting the closed-loop design.

Sensitivity to conflict metric. Table 4 compares cosine-, angle-, and weight-based conflict metrics on CIFAR-10 and CIFAR-100. No single metric dominates every dataset and partition, but the results remain close across met-

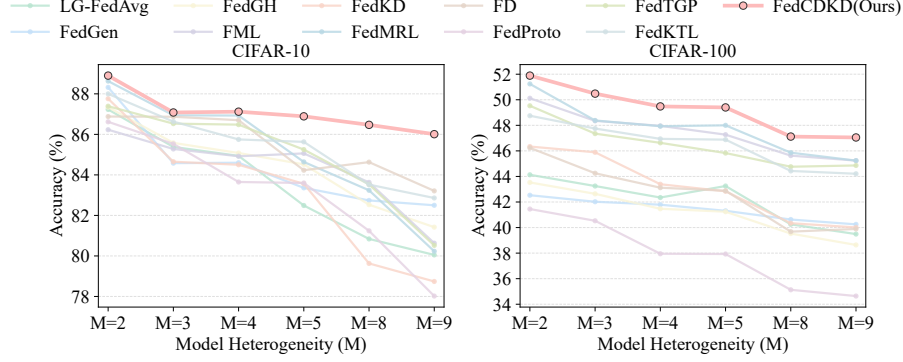


Fig. 4. Accuracy (%) under increasing model heterogeneity on CIFAR-10 and CIFAR-100 with $\text{Dir}(0.1)$. The horizontal axis varies the number of candidate client model families M ; larger M indicates a more diverse architecture pool and stronger representation mismatch across clients.

Table 4. Conflict-metric sensitivity on CIFAR-10 and CIFAR-100. Accuracy (%) is reported under Dirichlet splits with different concentration parameters.

Dataset	Metric	Dir(0.1)	Dir(0.3)	Dir(0.5)
CIFAR-10	Cosine	86.88	72.61	53.54
	Angle	87.04	72.35	53.96
	Weight entropy	87.12	72.44	53.89
CIFAR-100	Cosine	44.90	45.03	45.00
	Angle	45.41	45.78	44.77
	Weight entropy	45.32	45.06	45.05

rics. This indicates that FedCDKD benefits from conflict-aware control while the conclusion is not tied to one particular disagreement measure.

EMA coefficient. The supplementary material reports the sensitivity to the EMA direction-tracking coefficient s_u . FedCDKD is not highly sensitive over the tested range, which suggests that the method depends more on smoothed direction information than on a narrow value of s_u .

Resource usage. The supplementary material also reports architecture-level parameter and FLOP profiles for the visual backbones. FedCDKD does not require uploading private personalized models and adds only one scalar conflict degree per selected client. This keeps communication at the same order as standard shared-model FL, while allowing clients to keep deployment-specific architectures.

Communication and server overhead. FedCDKD adds only one scalar conflict degree per selected client to the usual shared-model broadcast, and clients upload only the shared model θ rather than the private personalized model ϕ . The main server-side overhead is the $O(K^2d)$ Gram matrix construction plus a simplex-constrained QP over K variables. Thus, FedCDKD preserves the communication order of standard FL while adding conflict-aware control.

We do not claim zero computational overhead; wall-clock server profiling under large-scale deployments remains future work.

Taken together, the results show that FedCDKD’s gains come from the complementarity between conflict-suppressed aggregation and conflict-driven distillation.

6 Conclusion

We studied heterogeneous federated visual recognition where model heterogeneity and data heterogeneity coexist with partial participation, often causing severe conflict among client update directions. To address this setting, we proposed FedCDKD, a conflict-driven knowledge distillation framework that couples server-side aggregation with client-side distillation in a closed loop. On the server, FedCDKD tracks client update directions with EMA and computes a stable consensus update by solving a min-norm QP, which suppresses conflicting components and produces a personalized conflict signal. On the client, this signal adapts the distillation temperature and loss weights, enabling bidirectional mutual distillation between a shared aggregatable model and a private personalized model. Experiments on visual benchmarks show that FedCDKD improves heterogeneous FL performance, especially under severe label skew, low participation rates, stronger model heterogeneity, and larger local update budgets.

Limitations and future work. Our evaluation follows simulated federated protocols, which are standard for controlled comparison but do not fully capture deployment factors such as unstable links, device failures, battery constraints, or privacy attacks on side-channel signals. We analyze communication order and server-side complexity, but do not report hardware-specific wall-clock profiling of the min-norm QP or full per-round energy cost. Future work will extend FedCDKD to more deployment-realistic visual FL settings and study formal privacy protection for the broadcast conflict signal.

Acknowledgement. This work was supported by the Heilongjiang Provincial Natural Science Foundation of China (Grant No. LH2024F035), and the funding from the Basic Research Support Program for Outstanding Young Teachers in Provincial Undergraduate Universities in Heilongjiang Province (Grant No. YQJH2023302).

References

1. McMahan, B., Moore, E., Ramage, D., Hampson, S., y Arcas, B.A.: Communication-efficient learning of deep networks from decentralized data. In: Artificial Intelligence and Statistics. Proceedings of Machine Learning Research, vol. 54, pp. 1273–1282. PMLR (2017)
2. Li, X., Huang, K., Yang, W., Wang, S., Zhang, Z.: On the convergence of fedavg on non-iid data. In: International Conference on Learning Representations (2020)

3. Li, T., Sahu, A.K., Zaheer, M., Sanjabi, M., Talwalkar, A., Smith, V.: Federated optimization in heterogeneous networks. *Proceedings of Machine Learning and Systems* **2**, 429–450 (2020)
4. Karimireddy, S.P., Kale, S., Mohri, M., Reddi, S., Stich, S., Suresh, A.T.: Scaffold: Stochastic controlled averaging for federated learning. In: *International Conference on Machine Learning. Proceedings of Machine Learning Research*, vol. 119, pp. 5132–5143. PMLR (2020)
5. Acar, D.A.E., Zhao, Y., Matas, R., Mattina, M., Whatmough, P., Saligrama, V.: Federated learning based on dynamic regularization. In: *International Conference on Learning Representations* (2021)
6. Ye, R., Xu, M., Wang, J., Xu, C., Chen, S., Wang, Y.: Feddisco: Federated learning with discrepancy-aware collaboration. In: *Proceedings of the 40th International Conference on Machine Learning. Proceedings of Machine Learning Research*, vol. 202, pp. 39879–39902. PMLR (2023)
7. Li, Z., Lin, T., Shang, X., Wu, C.: Revisiting weighted aggregation in federated learning with neural networks. In: *Proceedings of the 40th International Conference on Machine Learning. Proceedings of Machine Learning Research*, vol. 202, pp. 19767–19788. PMLR (2023)
8. Tan, Y., Long, G., Liu, L., Zhou, T., Lu, Q., Jiang, J., Zhang, C.: Fedproto: Federated prototype learning across heterogeneous clients. In: *Proceedings of the AAAI Conference on Artificial Intelligence*. vol. 36, pp. 8432–8440 (2022)
9. Yi, L., Wang, G., Liu, X., Shi, Z., Yu, H.: Fedgh: Heterogeneous federated learning with generalized global header. In: *Proceedings of the 31st ACM International Conference on Multimedia*. pp. 8686–8696 (2023)
10. Lin, T., Kong, L., Stich, S.U., Jaggi, M.: Ensemble distillation for robust model fusion in federated learning. *Advances in Neural Information Processing Systems* **33**, 2351–2363 (2020)
11. Zhang, J., Liu, Y., Hua, Y., Cao, J.: FedTGP: Trainable global prototypes with adaptive-margin-enhanced contrastive learning for data and model heterogeneity in federated learning. In: *Proceedings of the AAAI Conference on Artificial Intelligence*. vol. 38, pp. 16768–16776 (2024)
12. Wang, J., Liu, Q., Liang, H., Joshi, G., Poor, H.V.: Tackling the objective inconsistency problem in heterogeneous federated optimization. *Advances in Neural Information Processing Systems* **33**, 7611–7623 (2020)
13. Gao, L., Fu, H., Li, L., Chen, Y., Xu, M., Xu, C.Z.: Feddc: Federated learning with non-iid data via local drift decoupling and correction. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. pp. 10112–10121 (2022)
14. Shi, C., Zhao, H., Zhang, B., Zhou, M., Guo, D., Chang, Y.: Fedawa: Adaptive optimization of aggregation weights in federated learning using client vectors. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. pp. 30651–30660 (2025)
15. Rehman, Y.A.U., Gao, Y., de Gusmao, P.P.B., Alibeigi, M., Shen, J., Lane, N.D.: Ldawa: Layer-wise divergence aware weight aggregation in federated self-supervised visual representation learning. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. pp. 16464–16473 (2023)
16. Lee, S., Zhang, T., Avestimehr, A.S.: Layer-wise adaptive model aggregation for scalable federated learning. In: *Proceedings of the AAAI Conference on Artificial Intelligence*. vol. 37, pp. 8491–8499 (2023)

17. Zheng, H., Hu, Z., Yang, L., Zheng, M., Xu, A., Wang, B.: Fedcalm: Conflict-aware layer-wise mitigation for selective aggregation in deeper personalized federated learning. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 15444–15453 (2025)
18. Zhang, J., Liu, Y., Hua, Y., Cao, J.: An upload-efficient scheme for transferring knowledge from a server-side pre-trained generator to clients in heterogeneous federated learning. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 12109–12119 (2024)
19. Jeong, E., Oh, S., Kim, H., Park, J., Bennis, M., Kim, S.L.: Communication-efficient on-device machine learning: Federated distillation and augmentation under non-iid private data. arXiv preprint arXiv:1811.11479 (2018)
20. Zhu, Z., Hong, J., Zhou, J.: Data-free knowledge distillation for heterogeneous federated learning. In: Proceedings of the 38th International Conference on Machine Learning. Proceedings of Machine Learning Research, vol. 139, pp. 12878–12889. PMLR (2021)
21. Wu, C., Wu, F., Lyu, L., Huang, Y., Xie, X.: Communication-efficient federated learning via knowledge distillation. *Nature Communications* **13**(1), 2032 (2022)
22. Yi, L., Yu, H., Ren, C., Wang, G., Li, X., et al.: Federated model heterogeneous matryoshka representation learning. *Advances in Neural Information Processing Systems* **37**, 66431–66454 (2024)
23. Hinton, G., Vinyals, O., Dean, J.: Distilling the knowledge in a neural network. arXiv preprint arXiv:1503.02531 (2015)
24. Sener, O., Koltun, V.: Multi-task learning as multi-objective optimization. In: *Advances in Neural Information Processing Systems*. vol. 31 (2018)
25. Yu, T., Kumar, S., Gupta, A., Levine, S., Hausman, K., Finn, C.: Gradient surgery for multi-task learning. In: *Advances in Neural Information Processing Systems*. vol. 33 (2020)
26. Krizhevsky, A., Hinton, G.: Learning multiple layers of features from tiny images. Tech. rep., University of Toronto (2009)
27. Le, Y., Yang, X.: Tiny imagenet visual recognition challenge. CS231N Course Project, Stanford University (2015)
28. He, K., Zhang, X., Ren, S., Sun, J.: Deep residual learning for image recognition. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR). pp. 770–778 (2016)
29. Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., Chen, L.C.: Mobilenetv2: Inverted residuals and linear bottlenecks. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR). pp. 4510–4520 (2018)
30. Szegedy, C., Liu, W., Jia, Y., Sermanet, P., Reed, S., Anguelov, D., Erhan, D., Vanhoucke, V., Rabinovich, A.: Going deeper with convolutions. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR). pp. 1–9 (2015)
31. LeCun, Y., Bottou, L., Bengio, Y., Haffner, P.: Gradient-based learning applied to document recognition. *Proceedings of the IEEE* **86**(11), 2278–2324 (1998)